WCS Logger

The assistant in WCS PC host applications

About this document

Scope and purpose

This automation tool assists users in collecting all necessary information / data through Saleae Logic Analysers.

Intended audience

Users who adapt Infineon WCS USB Type-C solutions in PC HOST applications and frequently communicate with Infineon WCS support teams in issue debugging discussions.

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1 Introduction

Infineon WCS CCG devices are commonly used in PD port controller applications in systems like Notebook and Desktop computers, tablets, mobiles etc. In such cases, the system design typically includes a Host Processor or Embedded Controller (EC) that is responsible for the system policy decisions. The CCG controller can be connected to the EC and the CCG firmware allows the EC to monitor the status of the USB-PD ports, change configurations, perform firmware updates, and transparently interact with other USB-PD devices connected to CCG. All the capabilities are exercised through commands/responses transferred across the Host Processor Interface (HPI).

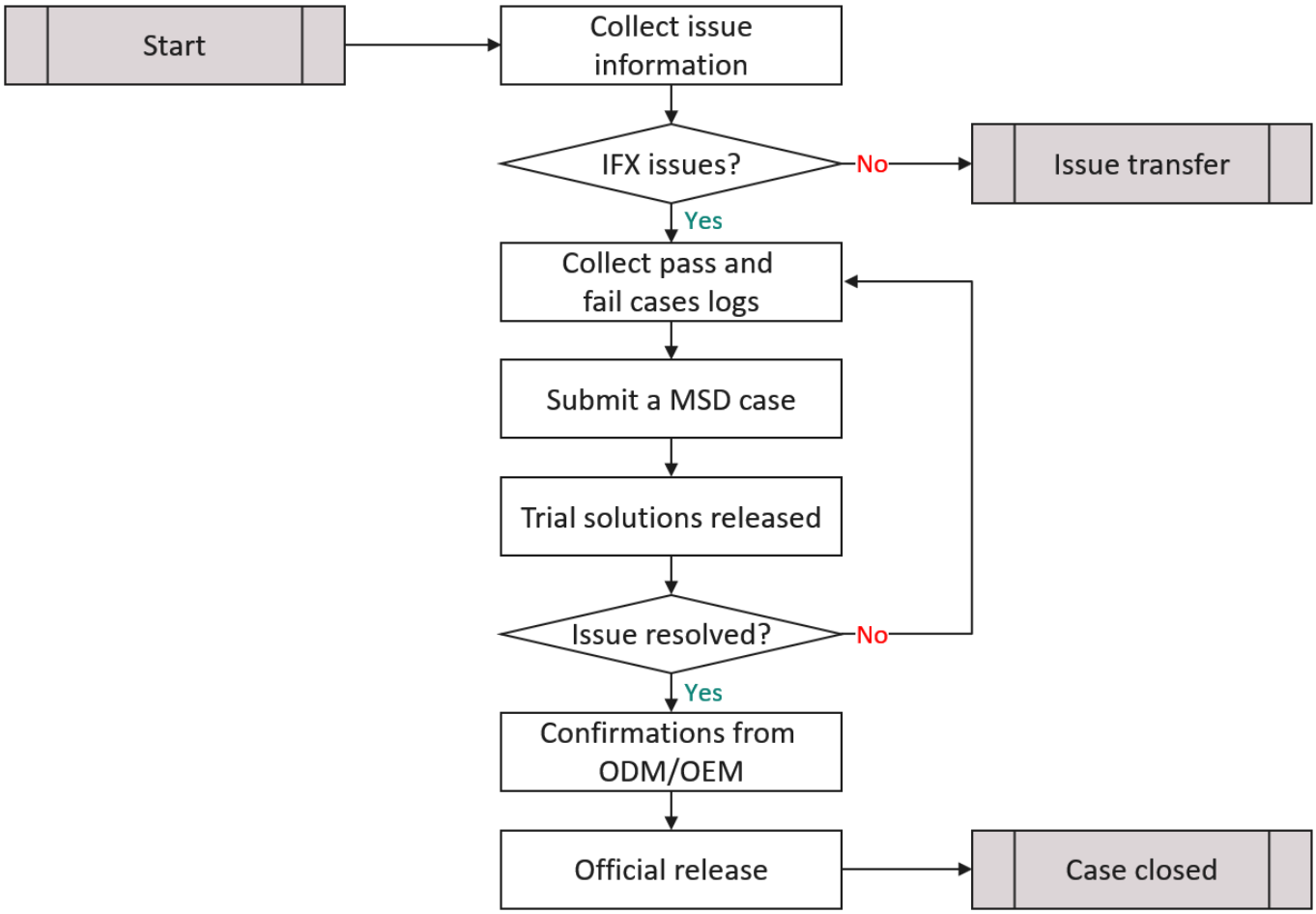


Figure 1 Standard issue report flow

1.1 Prerequisites

Infineon WCS support teams heavily rely on communication logs for issue analysis and debug, so a comprehensive log with accurate data is mandatory to accelerate solutions delivery. Saleae Logic Analyser is recommended as one of the prerequisites:

1. Saleae Logic Pro 16
2. Saleae Logic2 application v2.4.0 or above
3. Infineon CYPD4500 PD3.0 CC Sniffer
4. Infineon EZ-PD Protocol Analyser Utility

Introduction

Infineon WCS support team creates a tool package to facilitate and standardise log collection activities. It includes:

- **HPI Analyser**: a specialised I²C analyser dedicated to Infineon HPI protocol.
- **CC Analyser**: USB Type-C vendor define message analyser.
- **Infineon WCS Logger**: a python-base GUI tool to generate SALEAE logs with all necessary information described as WCS PD Host E2 training.

1.2 Installation

Few things ahead:

1. Install Saleae Logic2 application and enable "Automation" via Preference -> Automation -> Enable Automation Server.
2. Copy "***HPI_regs.xml***", "***i2c_analyzer_for_HPI.dll***" and "***Manchester_analyzer.dist.dll***" to the folder assigned in Custom Low Level Analysers.
3. Copy "***main_auto_logger.exe***" to arbitrary folder except root directory, which might need additional access permission for file creation later.

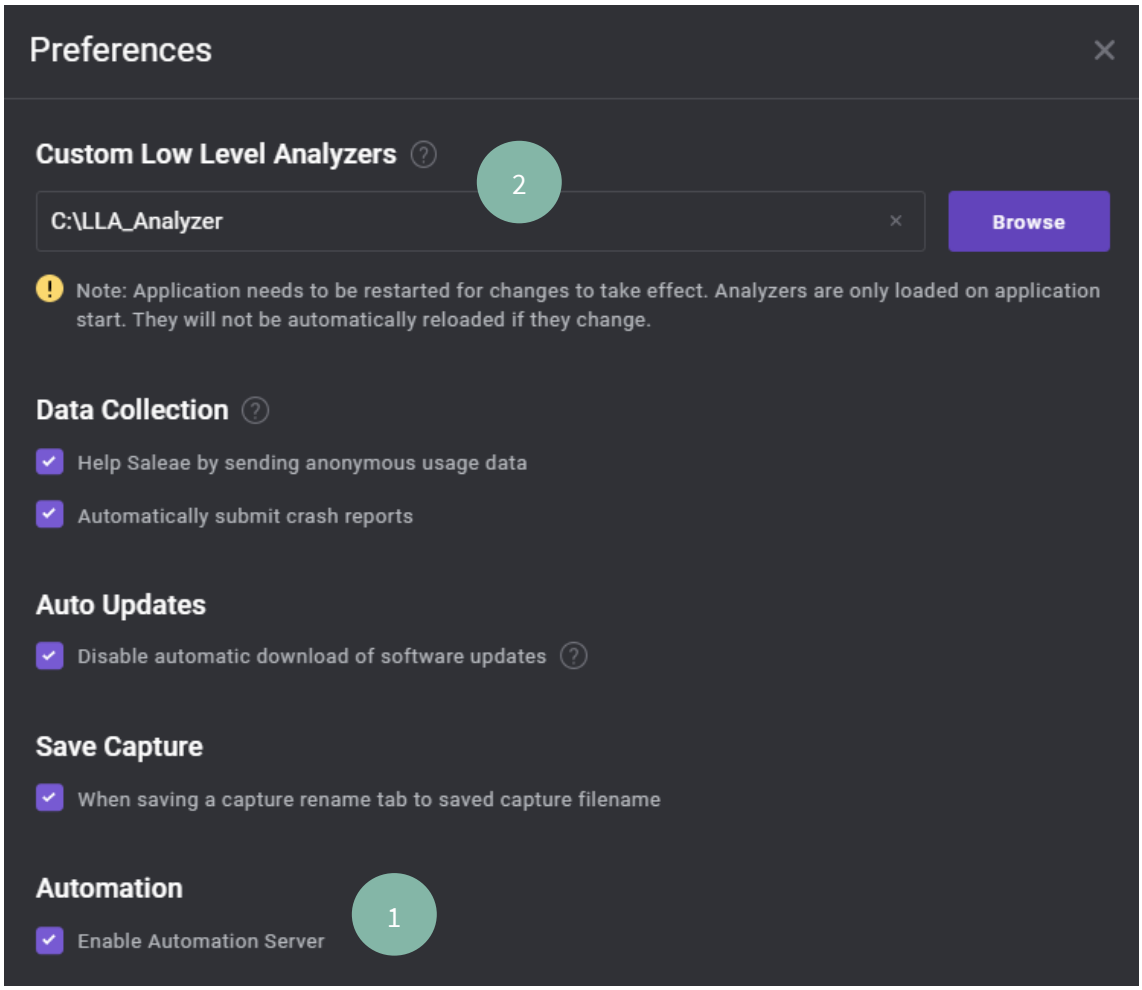


Figure 2 Saleae Logic2 Preferences

2 Infineon Automation Tool Package

2.1 HPI Analyser

The native I²C Analysers in Saleae Logic2 application can decode and list bus messages to its terminal-like window and tabular-format table for search usage. It decodes I²C commands in BYTE format basis and is useful in generic use cases.

However, Infineon HPI protocol usually conducts a WORD or longer data length as a single command and responses, and it is not intuitive to translate decoded messages through native I²C analyser. Worse, it is hard to do HPI command search. Sometimes it causes additional efforts and inconveniences to users, especially for engineers who are dealing with HPI activities in daily basis.

Hence, Infineon WCS support team creates a customised Saleae I²C analyser alternative. It can re-synthesise standard I²C byte packets to scalable-length message format, allows users to change significant byte order and capable to translate HPI raw data. In addition, it is easier to do HPI command analysis by introducing timestamp ahead of every HPI messages.

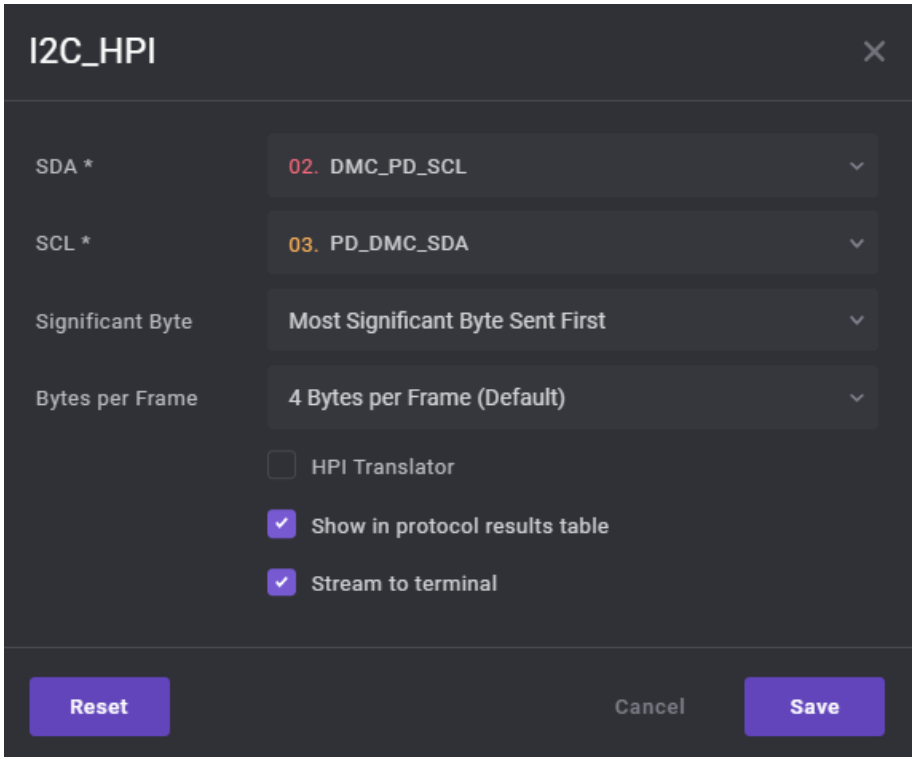
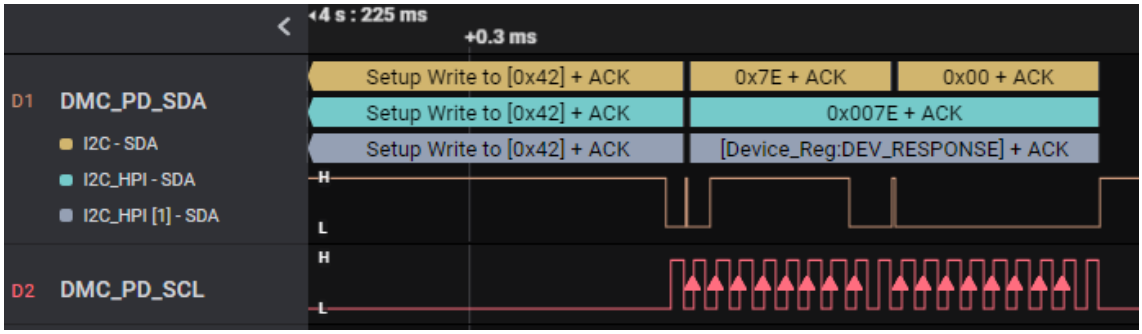


Figure 3 Additional options in Infineon "I2C_HPI" analyser



RESPONSE_REGISTER : 2 Bytes CCG1/CCG2 Address: 0x7E CCG3/CCG4/CCG5 Address: 0x007E			
Field	**Field Name**	**R/W[2]**	**Description**
Byte[0]	Response /Message Code	R	b7 – Type of Message 0 – Response to command issued by EC 1 – Event or Asynchronous Message b6:b0 – Message Code

Figure 4 HPI register address is decoded in raw (green) and translated (grey)

2.2 CC Analyser

USB CC and HPI logs are mandatory information WCS support team usually need while dealing with issues. In the past, they were provided by CPYD4500 CC sniffer and Saleae Logic Analyser separately, and always was difficult to understand command sequences among CC and HPI activities because these logs are not referring the same timeline.

By introducing Infineon CC Analyser with a proper configuration, users can capture and decode both USB CC and HPI logs through Saleae Logic2 application alone. It is extremely useful to check relevant command sequences across different protocols and accelerate debugging progress.

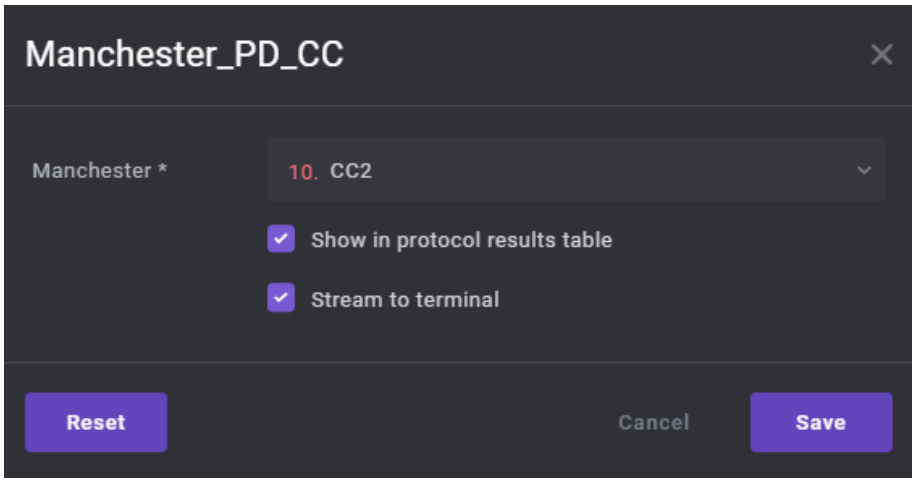


Figure 5 Infineon "Manchester_PD_CC" Analyser

2.3 Infineon WCS Logger

This is a portable application that supports Microsoft Windows 10 and above. It is highly recommended to use it with Saleae Logic Analyser Pro 16. The application pre-defines all necessary trace pins that users are required to provide while reporting issues for both AMD and INTEL design cases, along with project-relevant information, such as PD firmware revision. All numbers in pulldown menus are selectable but cannot be duplicated.

	INTEL Platform	AMD Platform	EVK Board	Checking Points
HW Signal	PCH_INT0# PCH_INT1# (Maple Ridge only) PCH_SDA PCH_SCL BBR_PWR BBR_RST SBU1/2	APU_RST# APU_INT APU_SDA APU_SCL Retimer_SDA Retimer_SCL HPD SBU1/2	MUX_SDA MUX_SCL HPD SBU1/2	Is the I/O behavior normal !? Is the SBU signal normal !?
CC Log	CC Log	CC Log	CC Log	Is enter DP Alt mode? Is device issue the Attention HPD High message to Host?
I2C Log	PCH I2C Log Retimer I2C Log	APU I2C Log Retimer I2C Log	MUX I2C Log	Compare the PASS log and Fail log to dig out is anything abnormal.

Figure 6 WCS PD Host E2 training

SALEAE log files will be generated based on channel number settings, associated with corresponding bus analysers respectively. These files are named beginning with DATE_TIME as a prefix and appended with selected data in the Version Information section. In addition, an issue tracker form and a text file named by the current date will be outputted under the same naming folder.

The Main window splits in following sections (refer to Figure 7):

1. **Version Information**: it is mandatory to fill all information correspondingly, whereas OEM Ticket# as optional. It also covers both AMD and INTEL design trace needs. All information will be recorded into issue tracker form automatically.
2. **Generic**: this section describes common trace pins need in WCS PC host application:
    a. USB-C: it defines SALEAE channel numbers to basic USB Type-C pins. All channels are recorded in digital mode as default, and can be set to digital + analogue mode by enabling the “Analogue Mode” option if necessary.
    b. PD/PD to EC: it defines HPI bus to EC and PD UART.
3. **AMD/INTEL Platform**: it defines all necessary trace pins need for AMD and INTEL designs respectively. Please be notices that due to insufficient pin counts available in SALEAE, in AMD designs, MUX power pin SALEAE channel is shared with HPD pin. Please connect either one depends on issue category for AMD design cases.
4. **Start/Abort Recording**: the main button to control Saleae Logic2 application. It will be grey-out if data missing in Version Information section or channel selection duplication. Use this button to start / stop log capturing processes.
5. **Pass/Fail Case**: the main buttons to save recorded logs for pass and fail cases. By clicking either buttons, it will save the current logs as pass or fail cases and export to an issue tracker form with all data described in Version Information section. It will be grey-out before log capturing.

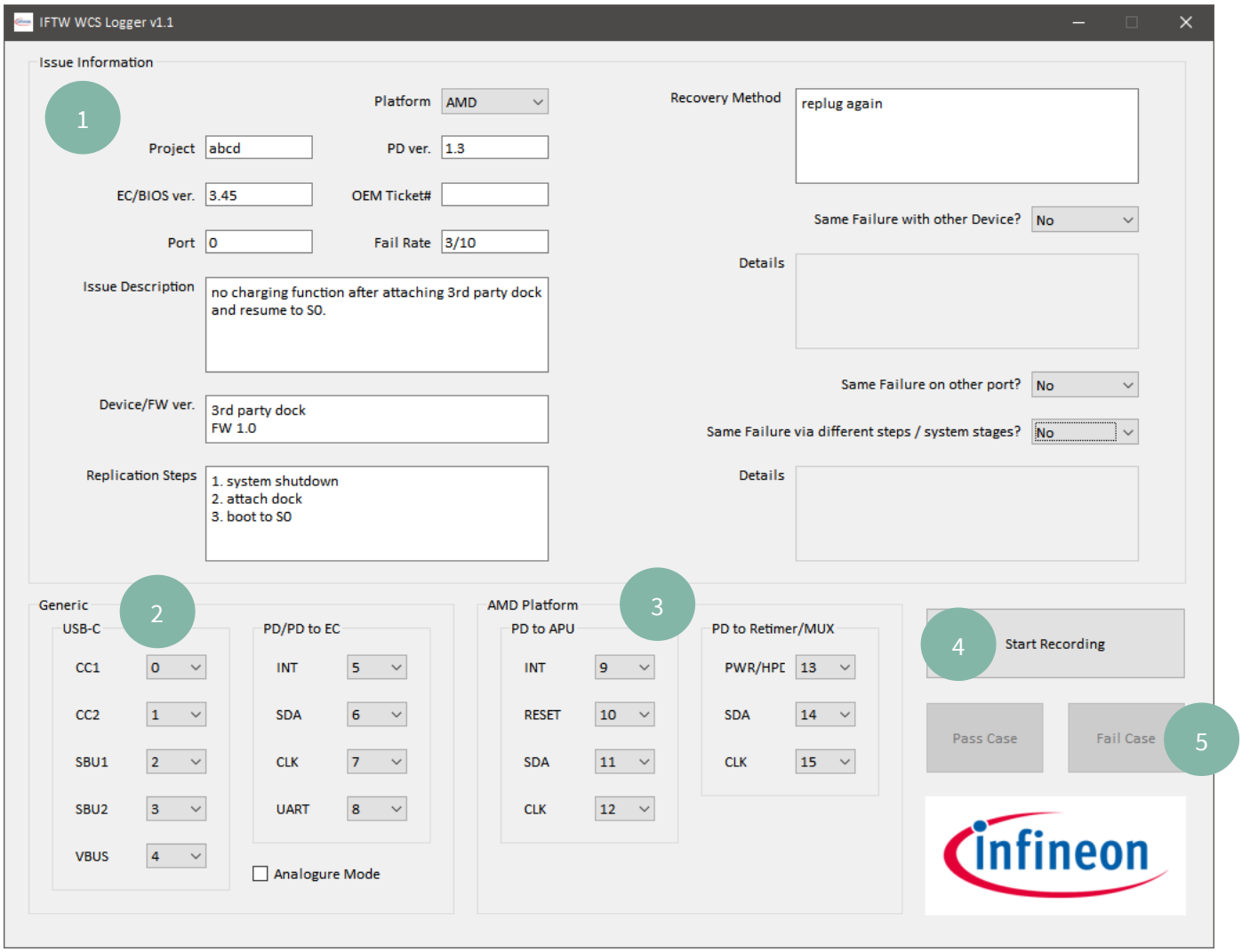


Figure 7 Infineon WCS Logger

3 How to use

Here is functional flow chart to illustrate Infineon WCS logger working model. Depends on Saleae Logic2 application running condition, there are 2 possible scenarios in Infineon WCS Logger usage:

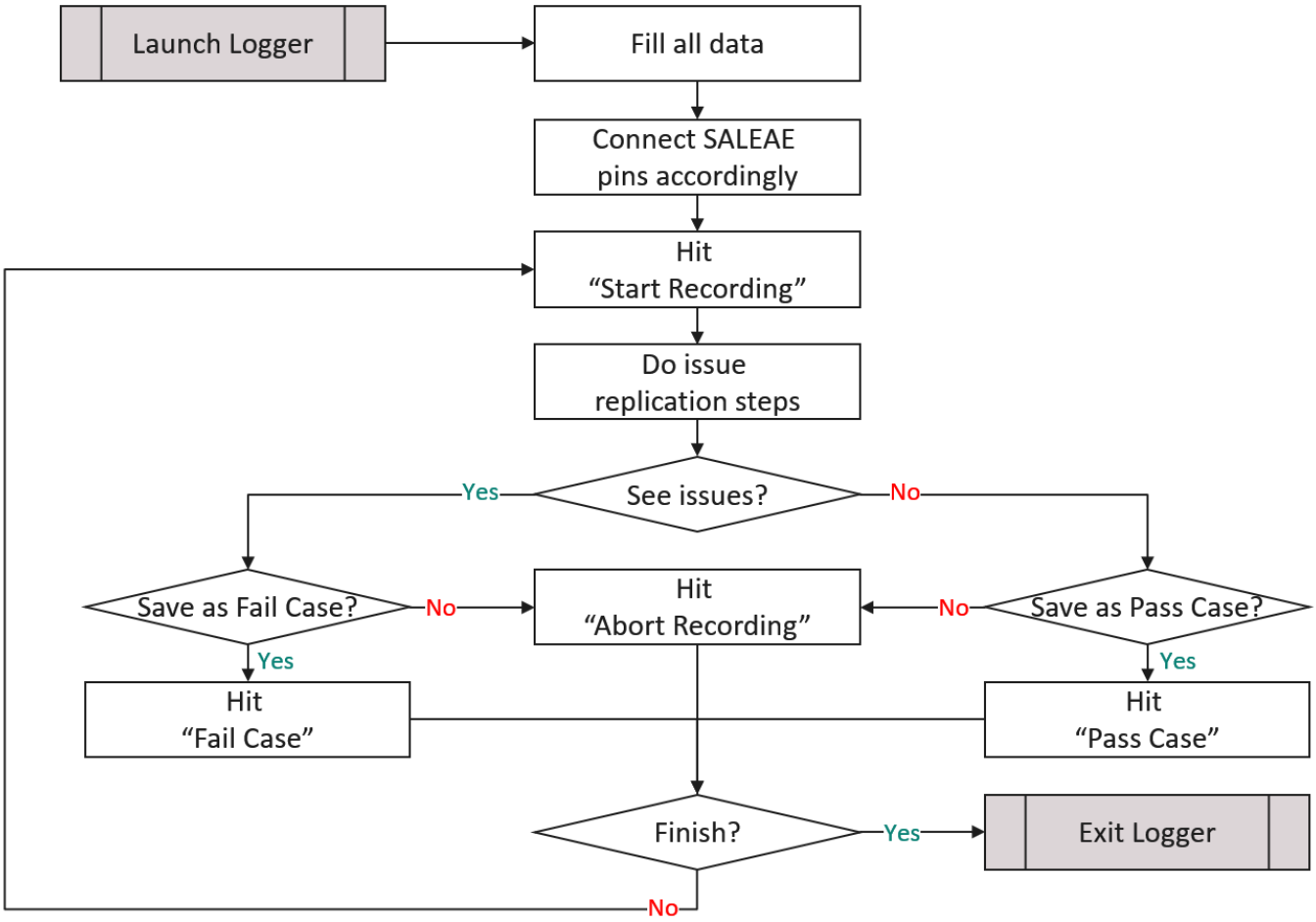


Figure 8 Infineon WCS Logger functional flow

3.1 Scenario A

This user scenario presumes Saleae Logic2 application is not running in the background while starting Infineon WCS Logger application. The benefit is users do not need to enable Saleae Automation option beforehand.

1. Start Infineon WCS Logger.
2. Fill all information and platform type in Version Information and mapping SALEAE pins with pre-defined channels or preferred ones.
3. Click "Start Recording" to start log capturing per setup.
4. Infineon WCS Logger launches Saleae Logic2 application with automation supportive.
5. Do issue replication steps. (e.g., shutdown and reboot to S0, re-plugging devices again, etc.)
6. Click "Abort Recording", "Pass Case" or "Fail Case" while replication steps finished.
7. Infineon WCS Logger assigns bus analysers to generated SALEAE log, and record capture settings in the text configuration file if necessary.

8. Repeat step 3 to 7 for another log capture round.
9. Close Infineon WCS Logger while finished, and Saleae Logic2 application will be shut down.

3.2 Scenario B

This user scenario only works while automation option is enabled in Saleae Logic2 application as pre-requisites. In this case, it is presumed users launch Saleae Logic2 application already.

1. Start Infineon WCS Logger.
2. Fill all information and platform type in Version Information and mapping SALEAE pins with pre-defined channels or preferred ones.
3. Click “Start Recording” to start log capturing per setup.
4. Do issue replication steps. (e.g., shutdown and reboot to S0, re-plugging devices again, etc.)
5. Click “Abort Recording”, “Pass Case” or “Fail Case” while replication steps finished.
6. Infineon WCS Logger assigns bus analysers to generated SALEAE log, and record capture settings in the text configuration file if necessary.
7. Repeat step 3 to 6 for another log capture round.
8. Close Infineon WCS Logger while finished, and Saleae Logic2 application will ******NOT****** be shut down.

4 Limitations of Infineon WCS Logger

There are several limitations of Infineon WCS Logger, including:

1. It supports Microsoft Windows 10 and above only.
2. It might crash if Saleae Logic2 application does not enable automation option while working at Scenario B.
3. Controlling Saleae Logic2 application directly might cause Infineon WCS Logger crash.
4. It does not support Saleae Logic Analyser 8 / Pro 8 due to insufficient channel numbers.
5. Channel renaming is not supported due to limitation of Saleae Automation library.

Self Service

CATEGORY	SYMPTOM	BASIC GUIDELINE
CHARGING	Charing icon is still ON after disconnecting USBC power source / no charging icon after connecting USBC power source	- Transfer to EC if relevant HPI commands (connect / disconnect) are sent from PD.
	System shows “PC isn’t charging” message while attaching USBC power source devices	- Transfer to power source devices if no source PDOs or insufficient power profile capability are broadcasted.
	System shows “charging slowly” in battery setting dialog while attaching USBC power source	-
DISPLAY	No display	- Transfer to graphic driver if EDID is shown in display setting. - Transfer to external monitor / dongle if no HPD high or IRQ on CC log.
	Display flicking / blur / flake on screen	- Transfer to graphic driver.
	System shows “Display connection might be limited” message while attaching USBC display accessory	- Transfer to display accessory.
USB	No device is shown in device manager	NEED USB MP310Q
	System shows “USB device not recognized” message while attaching USBC device	
	Device is enumerated but yellow bang in device manager	
THUNDERBOLT	No Thunderbolt function	- Check if INTEL ALT MODE is entered properly or not.

Revision history

Document version	Date of release	Description of changes
1.00	2023-07-20	First release.

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